AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 3: PROJECT DESIGN

1. INTRODUCTION

The Project Design Phase describes the complete design and structure of the Auto Ticket Classification using Flow Designer project. The purpose of this phase is to define how the proposed system will work before starting the actual development. The design provides a clear understanding of the different components involved in the system and explains how these components communicate with each other.

The proposed system uses Service Now Flow Designer to automate the classification of support tickets. When a new ticket is created, the system reads the available ticket information and checks it against predefined conditions. Based on the matching condition, the ticket is automatically classified and the required fields are updated.

2. SYSTEM DESIGN

The system is designed to reduce the manual effort involved in classifying support tickets. In a traditional process, a service desk agent needs to read each ticket, understand the issue, select the appropriate category, and assign the ticket to the relevant support team. This process can take additional time when a large number of tickets are received.

The proposed system automates these activities using Flow Designer. The flow monitors the creation of a new ticket and processes the ticket information according to predefined business rules. The system then identifies the appropriate classification and updates the ticket automatically.

The overall process can be represented as:

Ticket Created → Flow Trigger → Read Ticket Details → Check Conditions → Identify Classification → Update Ticket → Assign Group → End

3. SYSTEM ARCHITECTURE

The system architecture consists of several components that work together to perform automatic ticket classification. The user first creates a support ticket in Service Now. The ticket record stores important information such as the short description, description, category, priority, and other relevant details.

Once the ticket is created, the Flow Designer trigger starts the automated process. The flow reads the information available in the ticket and checks it against the predefined classification

conditions. When a matching condition is found, the flow performs the required action and updates the ticket.

The major components of the system are the **User, Incident/Ticket Table, Flow Designer, Conditions, Classification Logic, Assignment Group, and Updated Ticket**. These components work together to complete the classification process.

4. FLOW DESIGN

The Flow Designer is the main automation component of the project. It is responsible for controlling the complete workflow from the creation of a ticket to its final classification. The flow contains a trigger, conditions, and actions that determine how the ticket should be processed.

The flow starts when a new ticket is created. After the trigger is executed, the flow reads the relevant ticket information. The information is then checked against the conditions defined in the flow. If a condition is satisfied, the corresponding classification action is executed.

5. CLASSIFICATION LOGIC

The classification logic is an important part of the system design. It defines how the system identifies the type of ticket. The conditions are created based on the information available in the ticket.

For example, a ticket containing a laptop or desktop problem can be classified as **Hardware**. A ticket related to password or login problems can be classified as **Access**. Email-related problems can be classified as **Email**, while VPN and connectivity problems can be classified as **Network**.

The classification rules can be modified according to the requirements of the organization. This makes the system flexible and allows administrators to add new categories or modify existing conditions when required.

TICKET INFORMATION	CLASSIFICATION
Laptop/Desktop problem	Hardware
Password/Login problem	Access
Email problem	Email
VPN/Network problem	Network
Application/Software problem	Software

6. FLOW DESIGN STRUCTURE

The proposed Flow Designer structure begins with the creation of a new ticket. The trigger starts the flow and the ticket details are then evaluated. The flow uses conditional logic to determine the appropriate classification.

The flow can contain multiple **If/Else** conditions. Each condition checks for a particular type of issue. When the condition is true, the corresponding classification is selected and the ticket is updated. If none of the conditions are satisfied, the ticket can be kept as **Other** or **Unclassified** for manual review.

7. DATA FLOW DESIGN

The data flow explains how information moves through the system during the classification process. Initially, the user provides information while creating a ticket. This information is stored in the ServiceNow ticket record.

The Flow Designer accesses the ticket information and uses it as input for the classification conditions. The conditions analyze the available information and determine the appropriate classification. The result of the classification is then used to update the ticket fields.

The overall data movement can be represented as:

User Input → Ticket Record → Flow Designer → Condition Check → Classification → Ticket Update

This process ensures that the information provided by the user is used effectively for automatic classification.

8. TICKET UPDATE DESIGN

After identifying the ticket classification, the flow performs the required update operation. The appropriate category or classification field is updated based on the result of the condition check.

If the project requires assignment automation, the flow can also update the assignment group. For example, a network-related ticket can be routed to the network support team, while a hardware-related ticket can be routed to the hardware support team.

The ticket remains available in ServiceNow after the automated process is completed, allowing support agents to continue with the next steps required to resolve the issue.

9. USER INTERFACE DESIGN

The project uses the standard ServiceNow interface for ticket creation and management. The user does not need to interact directly with the Flow Designer during normal ticket creation.

The important ticket fields considered by the system may include **Short Description, Description, Category, Subcategory, Priority, Assignment Group, and Classification**.

The user enters the required ticket information, while the Flow Designer performs the classification automatically in the background. This makes the process simple for end users and reduces the amount of manual work required from service desk agents.

10. ERROR HANDLING DESIGN

The system should also consider situations where a ticket does not match any of the predefined conditions. Not every ticket may contain sufficient information for automatic classification.

In such cases, the flow should not assign an incorrect classification. Instead, the ticket can be marked as **Other** or **Unclassified** and made available for manual review by a service desk agent.

This approach helps prevent incorrect classification and ensures that tickets that require human attention are not automatically sent to the wrong support team.

11. SECURITY DESIGN

Security is an important consideration in the project design. Only authorized ServiceNow users should be allowed to create, modify, activate, or deactivate the Flow Designer workflow.

Access to ticket information and flow configuration should be controlled according to the organization's ServiceNow roles and permissions. This prevents unauthorized users from changing the classification logic or modifying the automation process.

12. EXPECTED DESIGN OUTPUT

The main output of the Project Design Phase is a complete technical design for the proposed automation. The design should clearly explain the system architecture, workflow, classification logic, data flow, ticket update process, error handling, and security requirements.

The completed design will provide a clear reference for implementing the actual Flow Designer workflow during the Project Development Phase.

13. CONCLUSION

The Project Design Phase provides a detailed blueprint for developing the Auto Ticket Classification system. It explains how the ticket information will be received, processed, classified, and updated using ServiceNow Flow Designer.

The design also provides a structured approach for handling different ticket types, unmatched tickets, assignment groups, security, and error conditions.